

Review of JPE_20241011 - "Self-Targeting in U.S. Transfer Programs"

This paper examines whether voluntary take-up of transfer programs leads individuals in greater need to receive the program than automatic enrollment, i.e. whether voluntary take-up induces beneficial self-selection on unobserved criteria. The paper first proposes a measure of (average) self-targeting to assess whether there is self-targeting and to characterize it. It then discusses measuring the value of self-targeting, distinguishing insurance value (stemming from individuals taking up the program at times of more need) and selection into who takes up the program. Finally, the paper develops a framework for the utility of marginal changes in the fraction of the program budget that is distributed automatically rather than through voluntary take-up. The authors use this framework to derive a sufficient statistic to evaluate marginal changes toward automatic enrollment. Each of these parts combines theory with a very carefully done application to the PSID.

Overall, the paper makes a very useful contribution to an important topic – how we should run the safety net and how to distribute government transfers efficiently. Both questions became very salient in the pandemic and apply to both emergency aid and the safety net. As classic allocation problems, they are more central to economics than many of the questions on program effects that receive much more attention. As such, I find the test and the evaluation framework very useful. My main concern, as discussed in the comments below, is that the paper leaves many important questions unclear. Some of it may just stem from the paper being dense and covering a lot of ground, but implementing the approach empirically still seems to require resolving several important questions.

Main Comments:

1. **Test for average self-targeting:** The coefficient on transfer receipt, β is the central statistic of the paper, so what it captures, how it should be estimated and what could go wrong should be very clear. The discussion is brief, hard to follow and some key issues are either skipped or found in other places. Some things just need to be explained more clearly or in the right place, here are two broad issues I would find important to clarify: First, it should be clear what it measures (average difference in the dependent variable between recipients and eligible non-recipients with the same income and X) and why $\hat{\beta} - \beta$ is a good statistic to test for self selection. Some specific issues that I found unclear are as follows: if I understand correctly, $\hat{\beta}$ only seems to measure the benefits of self-selection, but not the costs in terms of hassle and administration. So it can only tell us whether it can be worth it, we need an assessment of the costs to decide. I also think $\hat{\beta}$ captures a combination of self-selection and unmeasured eligibility criteria since some of those predicted eligible may be rejected applicants. Another issue I would like to see commented on is what kind of welfare criterion your test implies – as far as I understand, by weighting people equally, it ignores hassle costs and does not care about whether self-selection is more prevalent among the most needy or the marginal applicant. I believe these are reasonable choices and the test could be modified to reflect other choices by using weights in the regression.

A second important issue to discuss is your choice to use $\hat{\beta}$ rather than $\hat{\beta} - \beta$ as the test statistic. On p. 12 you transit from one to the other without even mentioning it. You later discuss behavioral responses, but that's much later and it is still not clear to me whether you assume that β is zero or whether you assume that $\beta > 0$, so that $\hat{\beta} < 0$ still implies advantageous self-selection. While I find your arguments regarding changes in consumption to be eligible reasonable, I think $\beta = 0$ is a very strong assumption. Behavioral changes in order to meet eligibility criteria do tend to be

small, but there could also be an effect of the transfer on consumption or a correlation between preferences for take-up and preferences for consumption that would lead to $\beta \neq 0$. This issue is amplified by the problem that one can never use a complete measure of consumption as the dependent variable, so that the assumption would also fail if program participation is associated with differences in consumption patterns or has an effect on consumption of measured vs. unmeasured consumption. That makes it a bit less clear whether β may not be negative, so it may not be as easy to infer advantageous self selection. Yet what is more important is that even if $\beta \geq 0$, the test becomes “one-sided” in that it can only detect positive self-selection, but cannot indicate a problem of negative self-selection. It also seems to make $\hat{\beta}$ of very limited use in the welfare analyses below.

There are also a few issues that are central to implementing and interpreting the test, that I think the paper should address. The first issue is the choice of dependent variable and its consequences. You construct it very carefully, but it would be useful to understand what the ideal measure would be (individual welfare/utility?), how feasible measures differ from that and what that implies as well as why your consumption-based measure is ideal among feasible choices. These questions are likely very similar to common questions of resource measures for poverty (raising questions such as how to deal with savings, investments, etc.) and measuring well-being by consumption (raising questions such as how measures are affected by shifts in consumption and being unable to measure all consumption). There is no need to reinvent that wheel, but since choosing the dependent variable is a key part of the test, readers should be aware of the relevant questions.

The second issue is the choice of independent variables. It would be good to briefly justify why you only control for income. I think that is sensible, but one could reasonably wonder whether it wouldn't be better to control for household structure rather than using equivalence scales, whether one should control for price levels (which will make the map from income to consumption vary across areas and could thus bias β). Yet the more important point is that I do not understand how you compare people with the same income when recipients have additional program income by definition. Does income include these transfers? If not, then there is a clear reason why recipients should have higher consumption than non-recipients (what else are they supposed to do with the transfer). Yet if you include it, we may be worried that the marginal propensity to consume out of a transfer differs from the marginal propensity to consume out of income, which would bias $\hat{\beta}$. That's clearly true for in-kind transfers such as housing, but could also be a problem especially if programs indeed provide insurance value (i.e. we'd expect those who have already played that card to save more). For in-kind transfers, errors in their valuation would also bias $\hat{\beta}$.

2. **Welfare Analysis:** I found it hard to follow some of the arguments in this part, maybe I am confused and it just needs to be explained (to me or in the paper). β seems to be something different in this part and its definition on p.25¹ does not seem to match the regression coefficient you use.² You say β is the difference between the marginal social values of income between

¹ As an aside, shouldn't the definition condition on eligibility?

² Note that if they are the same, the average and marginal value of self-targeting have to have the same sign (since variances are positive) and can only differ because of the second and third term, which does not have to be the case in practice (a program can be self-targeting on average, but become more self-targeting when the budget is reduced by a dollar).

recipients and non-recipients. Yet above, β is the average difference between consumption between the groups. I do not understand how a regression of consumption on program receipt is informative about the value of income, much less the *marginal* value of income. Unless you make very restrictive assumptions, how can the same regression give you the average utility in the test above and the marginal utility here?

I also struggled with some parts of the model setup and derivation. For example, your discussion of the optimal participation rate M on p. 30 suggests that you treat M as a choice parameter that can be chosen for a fixed elasticity and β . However, shouldn't both the elasticity of participation and the average self-selection change with the participation rate? If so, there is a derivative wrt. to these parameters that enters the optimization problem, so one cannot optimize M holding the other two terms fixed. So maybe I am misunderstanding key parts of the model.

I find the marginal analysis less compelling than the analysis of average self-selection for two additional reasons: First, in line with my struggles with β , the problem of analyzing consumption rather than utility seems more severe here. Utility is non-linear in consumption, so the marginal criterion will have an individual-level multiplicative factor that depends on the marginal utility of consumption of the recipients of the automatic vs. self-selected program. There is obviously little one can do about that, since we cannot measure utility. But mirroring my comment about welfare criteria in the average measure, I would expect some discussion of the consequences of replacing welfare weights with your consumption measure.

Second, I am not sure the analysis captures the most relevant policy consideration. It seems relevant for the question whether one or another transfer program should be expanded, which is indeed relevant. Yet the more frequent question is whether we should increase or decrease hassle costs or eligibility criteria of a given program. As you mention, that is a very difficult question, since we do not know who get screened out by specific hassle costs such as stigma. Yet even if the question is about re-locating money, an unattractive feature of your criterion is that the extensive margin of participation drops out in the marginal analysis. Unless I misunderstand, the criterion thereby captures the desirability of a reform where no one drops out of the self-selected program (since those who drop out are exactly indifferent, so they have point mass zero for continuous costs and benefits). By holding participation fixed, your criterion thereby implicitly seems to abstract from the very question you raise, which is the value of (changes in) self-targeting and instead evaluates whether adding a dollar to a given self-targeting regime offsets its costs (in terms of labor supply and hassle costs). In practice, the more relevant question seems to be how self-targeting changes with changes in the participation rate induced by changes in benefit amounts. For changes in benefit amounts that are not infinitesimal, whether those who drop out of the program are more or less self-selected clearly matters.³ The order in which people exit the program, which is difficult to infer, also affects whether the social welfare function is monotonic in the benefit amount. That we can neither assume monotonicity nor infer how β changes with M seems to make the sufficient statistic approach less attractive, since it can point away from the global optimum in these situations.

³ And even if they are indifferent in terms of balancing hassle costs and benefits of program participation – shouldn't our valuation of a marginal change depend on whether the marginal individual is positively or negatively self-selected (in addition to the inframarginal dollar that your formula captures)? I guess I am somewhat puzzled by the fact that the derivative of β with respect to M drops out of the equation.

3. **Implementation and Data:** The application is very carefully done and there is a lot of attention to detail, but it glosses over aspects of key relevance for its implementation. Some key choices do not seem to be explained. For example, you present a theory about income and consumption levels, but then jump to using income and consumption ranks in the application. At a minimum, that requires an explanation, but I think you need to use levels, since I do not see a useful welfare criterion that would allow you to use ranks. Another key issue in implementation is the prediction of eligibility. As a simple practical matter, you should discuss how you deal with recipients whom you predict to be ineligible. Simply excluding them raises several conceptual problems (predictors of eligibility likely negatively predict consumption and your robustness check only partly accounts for that) and implies that you cannot say anything about self-targeting of the program, since you are only able to analyze a non-randomly selected subpopulation.

But there also seem to be problems that are harder to address, since they stem from the fact that crucial information is not available or measured with substantial error. As you mention, predicting eligibility involves considerable error and it would be useful to discuss its extent and potential consequences/solutions. Without better data on eligibility, the extent of error in predicting eligibility (see Newman and Scherpf, 2013⁴ for some evidence) seems to make implementation of the test very difficult. I do not understand your argument regarding the direction of this bias in footnote 15. It seems to be based on the idea of OVB, yet misclassification of eligibility induces a sample selection problem. So the bias depends on expected consumption in the 4 relevant groups (2 correctly classified and 2 misclassified groups).

Another issue is the measurement of income. As far as I understand, you use annual income both in the analyses and in the determination of eligibility. Yet program eligibility typically depends on monthly/current income. On the negative side, that leads to a lot of misclassification in eligibility. On the positive side, it means that your estimates of the insurance value omit the value of insurance against within-year variation in income, so that the insurance value of these transfers is even more impressive. In a similar vein, I am not sure how to think about your analysis given that participation is also monthly rather than annual. My impression is that one would need (much more accurate) measures of variables such as income, consumption, eligibility and program participation at the monthly level to implement the test. And while I follow your arguments regarding survey error in program participation (though see minor point below), your arguments regarding measurement error in income do not address the most salient point: Even classical measurement error in income would lead to a negative bias in $\hat{\beta}$ (the coefficient on income is attenuated and income is negatively correlated with program participation, leading to negative bias), which could explain your results.

Some of the points above clearly arise from data limitation and you do an impressive amount of work to address them. I would prefer the paper to discuss these remaining shortcomings and what kind of data/methods would be required to address them, because doing so would allow researchers to improve and better apply the very useful test you propose.

Minor Points

⁴ Newman, Constance, and Erik Scherpf. 2013. "Supplemental Nutrition Assistance Program (SNAP) Access at the State and County Levels: Evidence from Texas SNAP Administrative Records and the American Community Survey." USDA Economic Research Service Working Paper ERR-156.

1. Misreporting of program receipt is a key problem for your test. You clearly point this out and do a lot to address it. While I share your conclusion, I am not sure you make the most convincing case for the validity of your test in the presence of misclassification. After all, several papers have demonstrated misclassification rates high enough to overturn your results. However, the first order effect of misclassification should be to attenuate your test statistic, simply because any random misclassification will make recipients and non-recipients more similar. This tendency could be overturned if misclassification is correlated with consumption. My intuition is that it would require a peculiar setup for this to generate spurious results - those who do not report true receipt need to have higher consumption than those who truly report receipt (which is plausible) as well as than eligible non-participants (which seems less likely).
2. Isn't a slightly odd feature of the adjustment to automatic enrollment that your proposed policy alternative punishes people at income levels with more self-selection more? That suggests that it may not be the welfare-maximizing adjustment. Maybe it is still the most reasonable adjustment, but it is a policy that seems unlikely to happen in practice, so it may be useful to have a short discussion of how these choices affect welfare analyses and how one could develop similar criteria for specific policy proposals.
3. P.22: I think throughout this discussion, "reducing" and "increasing" progressivity are flipped. Doesn't reducing progressivity mean that transfers increase less with income? So going automatic increases transfer progressivity, since it moves receipt higher up the income distribution? Just harmless wording.
4. Given the empirical problems, I find the conclusions of the empirical application to be stated rather strongly in some cases. They clearly point in one direction, but given the data and implementation issues above, I am less confident about conclusions regarding specific programs, how they compare and what the ideal policy would be than you seem to be. I am not sure whether and to what extent it is necessary to rephrase things more cautiously, but I see the application more as a proof of concept and usability (including proof of how much we need to improve the data) than as something I would use for policy here and now.